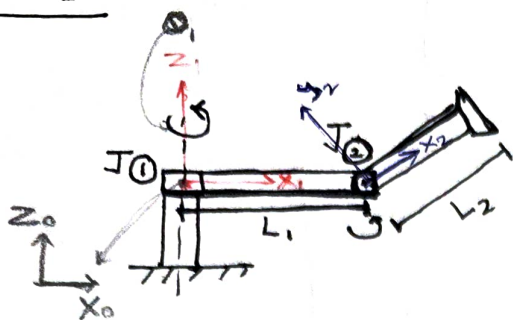


Ans 1



θ_i = The angle between x_i & x_{i-1}

d_i = Measured along z axis translation.
(offset of links along z -axis).
distance between θ_i & θ_{i-1} along the z axis. (the common ~~axis~~) by definition.

a_i = Length of Link or the length of common normal (length of x_i Link)

κ_i = The angle z_{i-1} need to twist in order to be parallel along z_i (Along x_{i-1})

DH Parameter table

i	a_i	κ_i	d_i	θ_i
1	$a_1 = L_1$	$\kappa_1 = 0$	$d_1 = 0$	θ_1
2	$a_2 = L_2$	$\kappa_2 = 90$	$d_2 = 0$	θ_2

Table

Homogeneous Transformation Matrix

$$T_i^{i-1} = \begin{bmatrix} R_i^{i-1} & P_i^{i-1} \\ 0 & 1 \end{bmatrix} = R_z(\theta_i) T_z(d_i) T_x(a_i) R_x(\kappa_i)$$

Rotation & translation about current axis

$$T_i^{i-1} = \begin{bmatrix} \cos \theta_i & -\sin \theta_i \cos \kappa_i & \sin \theta_i \sin \kappa_i & a_i \cos \theta_i \\ \sin \theta_i & \cos \theta_i \cos \kappa_i & -\cos \theta_i \sin \kappa_i & a_i \sin \theta_i \\ 0 & \sin \kappa_i & \cos \kappa_i & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T_2^0 = T_1^0 T_2^1$$

* Plugging in values of a_i κ_i d_i into T_1 & T_2 from Table 1

$$T_1^0 = \begin{bmatrix} \cos \theta_1 & 0 & \sin \theta_1 & L_1 \cos \theta_1 \\ \sin \theta_1 & 0 & -\cos \theta_1 & L_1 \sin \theta_1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T_2^1 = \begin{bmatrix} \cos \theta_2 & -\sin \theta_2 & 0 & L_2 \cos \theta_2 \\ \sin \theta_2 & \cos \theta_2 & 0 & L_2 \sin \theta_2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T_2^0 = T_1^0 T_2^1 = \begin{bmatrix} \cos \theta_1 \cos \theta_2 & -\cos \theta_1 \sin \theta_2 & \sin \theta_1 & L_1 \cos \theta_1 + L_2 \cos \theta_2 \cos \theta_1 \\ \sin \theta_1 \cos \theta_2 & -\sin \theta_1 \sin \theta_2 & -\cos \theta_1 & L_1 \sin \theta_1 + L_2 \sin \theta_2 \cos \theta_1 \\ \sin \theta_2 & \cos \theta_2 & 0 & L_2 \sin \theta_2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

P_x
 P_y
 P_z
 1

$$P_x = L_2 \cos \theta_2 \cos \theta_1 + L_1 \cos \theta_1$$

$$P_y = L_2 \sin \theta_2 \cos \theta_1 + L_1 \sin \theta_1$$

$$P_z = L_2 \sin \theta_2$$

Where $\{P_x, P_y, P_z\}$ is showing the position vector of end effector in base frame.